GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 6

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Recap of Previous Lecture

Topic Propositionlogic

Topic

Topic

Topic

Topic

Turn on Slide map

Topics to be Covered

- Topic — Predicate logic
- Topic
- Topic
- Topic
- Topic

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Propositionlogic :→

- Any Proposition can be T/F only.
- There was no variable

Predicate Logic

- Predicate logic, also known as **first-order logic (FOL)**, is an extension of propositional logic that deals with predicates, quantifiers, and variables.
- Predicates are properties, additional information to better express the subject of the sentence.
- A quantified predicate is a proposition , that is, when you assign values to a predicate with variables it can be made a proposition.

Predicate Logic

- In P(x) : x>5, x is the subject or the variable and '>5' is the predicate.
- P(7) : 7>5 is a proposition where we are assigning values to the variable x, and it has a truth value, i.e. True.
- The set of values that the variables of the predicate can assume is called the Universe or Domain of Discourse or Domain of Predicate.

P: >5 → Predicate

P(x) : x>5 → Can be true | false

x=7

- P(7)= 7>5 ⇒ True ⇒ Proposition
- In predicate if we assign value to x then ⇒ proposition.

Predicate Logic

- Predicates
- A predicate is a statement that contains variables and becomes a proposition when specific values are substituted for those variables. Predicates express properties or relations among objects.
- Example: P(x) = "x is an even number"
- When x=2, P(2) is True. When x=3, P(3) is False.

x is Subject
P: is an even number.

Predicate Logic

- Propositions can be True/ False.
- Example of Predicate : Ram is a Sportsperson. ← example

This statement Can be T/False

Here (Ram) is the subject.

(is a Sportsperson) is the predicate.

This is written as follows: P: is a sportsperson. → Predicate

P(x) : x is a sportsperson

$P(x)$,

Predicate applied on x

The value 'x' can take are called Universe, Domain of Predicate, Domain of discourse.

$P(Ram)$ = Ram is a sportsperson

Predicate applied on Ram

Predicate Logic

- One more example:

P : is a politician.

P(Sonu) : Sonu is a politician ⇒ this Statement Can be T/F

Subject

Predicate

Predicate Logic

- Example : using logical connectives.

Two predicate

- P: is an Actor
- S: is a Superstar

- $P(x) \rightarrow S(x)$

- if x is an Actor it implies x is a Superstar
- This will be true if P(x) is false, and P(x), S(x) are true.

Predicate Logic

- Example : using logical connectives.

- $P(x) \wedge S(x) \Rightarrow$ x is an actor and Superstar
  → True when both P(x), S(x) are true.

- $P(x) \vee S(x) \Rightarrow$ x is an actor or Super star
  → This will be true when any of P(x), S(x) are true

Predicate Logic

- We can have more than one variable in Predicate logic

Example ⇒

→ G(x, y) ⇒ x is greater than y.

→ P(x, y) ⇒ x is a Parent of y.

⇒ G(6, 5) ⇒ 6 is greater than 5 (T)

Predicate Logic

- Here we use quantifiers to express the extent to which a predicate is true.
- Quantifiers:
- Universal Quantifier (∀): This indicates that the predicate holds for all elements in the domain. Example: ∀x P(x) means "For all x, P(x) is true."
- Existential Quantifier (∃): This indicates that there exists at least one element in the domain for which the predicate holds. Example: ∃x P(x) means "There exists an x such that P(x) is true."

$\forall_x P(x) \Rightarrow$ This means that this predicate is True for all x in domain OR Tautology OR false for no x in domain

(Predicate applied on x)

$\exists_x P(x) \Rightarrow$ This means that this predicate is True for not all x in domain or for some x in domain or for at least one x in domain

Predicate Logic

- Universal Quantifier:

imp

- $\forall x P(x)$ → True , if domain of x is empty then
  $\forall x P(x) \Rightarrow$ definition: false for no x → True
- $\exists x P(x)$ → false , if domain of x is empty,
  $\exists x P(x) \Rightarrow$ definition for at least one x the P(x) shd be True. → false

Predicate Logic

- Universal Quantifier:

Predicate Logic

- Universal Quantifier:

Predicate Logic

- Existential Quantifier:

Predicate Logic

- Existential Quantifier:

Predicate Logic

- Existential Quantifier:

Predicate Logic

Let R(x) be the predicate "$x^2 \geq 0$" where x is a real number.

$R(x) \Rightarrow x^2 \geq 0$, domain

So $\forall x\, R(x)$

Let S(x) be the predicate "$x^2 = 4$" where x is a real number.

$S(x) \Rightarrow x^2 = 4$, $\exists_x S(x)$

Consider the predicate P(x,y): "x + y = 0" where x and y are integers.

$\Rightarrow \underbrace{\exists_x \exists_y P(x,y)}$

Predicate Logic

Let Q(x) be the predicate "x is prime" where x is a positive integer.

$\exists_x Q(x)$

domain

Let R(x,y) be the predicate "x < y" where x and y are real numbers.

$\exists_x \exists_y P(x,y)$

Problem 8: Let P(x) be the predicate "x is divisible by 3" where x is an integer. Determine the truth value of ∀x P(x) and ∃x P(x).

False

True.

Sab Kuch

Single Value

- $\forall_x \Rightarrow$ we have to check Truth for all x in domain
- $\exists_x \Rightarrow$ Check truth for at least one value

Predicate Logic

Unsolved Problems on Predicates and Quantifiers

1. Let P(x) be the predicate "$x^2 - 1 = 0$" where x is a real number. Determine the truth value of $\exists x\, P(x)$.

domain → True

- 2. Let Q(x,y) be the predicate "x < y" where x and y are integers. What does $\forall x\, \exists y\, Q(x,y)$ mean in words?

$Q(x,y) \Rightarrow (x<y)$

$\forall x \exists y\, Q(x,y)$

$\forall x$ → for all integer x

$\exists y$ → at least one y

False

$\forall x \forall y\, Q(x,y)$

$\exists x \exists y\, Q(x,y)$

$\exists y$ → at least one y.

$\exists x$ → at least one x

$\Rightarrow$ false

$\forall x \exists y \, (x<y)$

Randomly choose single y

$y=8$

Now check for all x

$x<8$ for all x

No

$\Rightarrow$ True

$\exists y \exists x \, (x<y)$

Choose single $x=4$

Can I choose any single y

that satisfy $x<y$, Yes $y=6$

Quantifier are applied on variable Ya Predicate

$\forall x \, P(x)$

Predicate Logic

Unsolved Problems on Predicates and Quantifiers

1. Let P(x) be the predicate "$x^2 - 1 = 0$" where x is a real number. Determine the truth value of $\exists x\, P(x)$. → True

domain

2. Let Q(x,y) be the predicate "x < y" where x and y are integers. What does $\forall x\, \exists y\, Q(x,y)$ mean in words? → False

3. Let R(x) be the predicate "x is even" where x is an integer. Write the statement "All integers are even" using predicate logic. → $\forall x R(x)$

4. Let S(x) be the predicate "x is a mammal" and T(x) be "x can fly" where x is an animal. How would you express "Some mammals can fly" using predicate logic?

x is an animal

$\Rightarrow \begin{cases} S(x): x \text{ is a mammal} \\ T(x): x \text{ Can fly} \end{cases}$

$\underbrace{\exists x}_{\text{There are Some animals}} \left(\underbrace{S(x)}_{\text{which are mammals}} \text{ and } \underbrace{T(x)}_{\text{Can fly}} \right) \Rightarrow \exists x \left(S(x) \wedge T(x) \right)$

Predicate Logic

Unsolved Problems on Predicates and Quantifiers

1. Let P(x) be the predicate "$x^2 - 1 = 0$" where x is a real number. Determine the truth value of $\exists x\, P(x)$. → True
   domain
2. Let Q(x,y) be the predicate "x < y" where x and y are integers. What does $\forall x\, \exists y\, Q(x,y)$ mean in words? → False
3. Let R(x) be the predicate "x is even" where x is an integer. Write the statement "All integers are even" using predicate logic. → $\forall x R(x)$
4. Let S(x) be the predicate "x is a mammal" and T(x) be "x can fly" where x is an animal. How would you express "Some mammals can fly" using predicate logic?
5. Let P(x,y) be the predicate "x is the parent of y" where x and y are people. Express "Everyone has a parent" using predicate logic.
   → (For all y there exist x)

- x, y are people
- x is the parent of y →

Everyone has Parent :-

a) $\forall x \forall y P(x,y)$
b) $\forall x \exists y P(x,y)$
c) $\forall y \exists x P(x,y)$
d) $\exists x \exists y P(x,y)$

Predicate Logic

Unsolved Problems on Predicates and Quantifiers

1. Let P(x) be the predicate "$x^2 - 1 = 0$" where x is a real number. Determine the truth value of $\exists x\, P(x)$. → True
   domain
2. Let Q(x,y) be the predicate "x < y" where x and y are integers. What does $\forall x\, \exists y\, Q(x,y)$ mean in words? → False
3. Let R(x) be the predicate "x is even" where x is an integer. Write the statement "All integers are even" using predicate logic. → $\forall x R(x)$
4. Let S(x) be the predicate "x is a mammal" and T(x) be "x can fly" where x is an animal. How would you express "Some mammals can fly" using predicate logic?
5. Let P(x,y) be the predicate "x is the parent of y" where x and y are people. Express "Everyone has a parent" using predicate logic.
6. Let Q(x) be the predicate "x is prime" where x is a positive integer. What is the negation of $\forall x\, Q(x)$? Skip
7. Let R(x,y) be the predicate "x + y = 10" where x and y are integers. Is the statement $\forall x\, \forall y\, R(x,y)$ true or false? → False
   Explain why.
8. Let S(x) be the predicate "x is a multiple of 5" where x is an integer. Express the statement "There is no integer that is a multiple of 5" using predicate logic.

a) $\forall x\, S(x)$ b) $\exists x\, S(x)$ c) $\sim \forall x\, S(x)$ d) $\sim \exists x\, S(x)$

Predicate Logic

Unsolved Problems on Predicates and Quantifiers

1. Let P(x) be the predicate "$x^2 - 1 = 0$" where x is a real number. Determine the truth value of ∃x P(x). → True (domain)
2. Let Q(x,y) be the predicate "x < y" where x and y are integers. What does ∀x ∃y Q(x,y) mean in words? → False
3. Let R(x) be the predicate "x is even" where x is an integer. Write the statement "All integers are even" using predicate logic. → $\forall x R(x)$
4. Let S(x) be the predicate "x is a mammal" and T(x) be "x can fly" where x is an animal. How would you express "Some mammals can fly" using predicate logic?
5. Let P(x,y) be the predicate "x is the parent of y" where x and y are people. Express "Everyone has a parent" using predicate logic.
6. Let Q(x) be the predicate "x is prime" where x is a positive integer. What is the negation of ∀x Q(x)? Skip
7. Let R(x,y) be the predicate "x + y = 10" where x and y are integers. Is the statement ∀x ∀y R(x,y) true or false? Explain why. → False
8. Let S(x) be the predicate "x is a multiple of 5" where x is an integer. Express the statement "There is no integer that is a multiple of 5" using predicate logic.
9. Let T(x,y) be the predicate "x is taller than y" where x and y are people. Express the statement "Nobody is taller than themselves" using predicate logic.
10. Let P(x) be the predicate "x is a planet in our solar system" where x is a celestial body. How would you express "There are exactly 8 planets in our solar system" using predicate logic?

- $\forall x P(x) \Rightarrow$ P(x) is true for all x | Nowhere P(x) is false for any x.
- $\sim \forall x P(x) \Rightarrow$ P(x) is not true for all x | It is false for at least once.
- $\exists x P(x) \Rightarrow$ P(x) is true at least for one value of x | P(x) is not false for all x
- $\sim \exists x P(x) \Rightarrow$ P(x) is not true for any x | P(x) is false everywhere

- $\forall x P(x) \Rightarrow$ P(x) is True all x | P(x) is not false for any x
- $\sim \forall x P(x) \Rightarrow$ P(x) is not True for all x | P(x) is false at least once

P(x) is true for some x

P(x) is not false for some x

- $\exists x P(x) \Rightarrow$ P(x) is true at least once | P(x) is not false for all x
- $\sim \exists x P(x) \Rightarrow$ P(x) is true for no x | P(x) is false for all x

Predicate Logic

- The part of the logical expression to which a quantifier can be applied is called the scope of the quantifier.
- A variable whose occurrence is bounded by a quantifier is called a bounded variable. Variables not bounded by any quantifiers are called free variables.

Predicate Logic

. **Examples of Predicate Logic Statements**

- **Universal Example**:

  `∀x (Human(x) → Mortal(x))`

  Translation: "For all `x`, if `x` is a human, then `x` is mortal."

- **Existential Example**:

  `∃x (Human(x) ∧ Rich(x))`

  Translation: "There exists an `x` such that `x` is both human and rich."

- **More Complex Example**:

  `∀x ∃y (Parent(y, x) ∧ Loves(y, x))`

  Translation: "For every `x`, there exists a `y` such that `y` is a parent of `x` and `y` loves `x`."

Predicate Logic

THANK - YOU